

Designing Constrained Generative NPCs with a Dynamic Evidence Graph for Investigative Narrative Worlds

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Abstract

Large language models are increasingly used to generate adaptive dialogue for narrative games, but unconstrained generation often introduces logical inconsistencies and premature disclosure of narrative-critical information. These issues are particularly problematic in investigative settings, where player reasoning depends on structured evidence and controlled information flow.

This poster presents a constrained generative NPC system that explores an alternative to semantic similarity-based retrieval. Narrative knowledge is organized as a Dynamic Evidence Graph (DEG) composed of interdependent Knowledge Fragments with explicit states, dependencies, trust thresholds, and spatial conditions. An engine-controlled Eligibility and Gating mechanism selects which fragments may be expressed at runtime, producing a constrained context for dialogue generation.

We illustrate the approach through a 2D investigative game prototype and formative expert feedback. The work highlights how explicit evidence structures and deterministic knowledge access can support coherent and inspectable generative NPC behavior in investigative narratives.

CCS Concepts

• **Human-centered computing** → **Interactive narrative**; *Natural language interfaces*; *User interface design*; • **Computing methodologies** → **Artificial intelligence**.

Keywords

Interactive narrative, generative NPCs, constrained generation, investigative games

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1 Introduction

Large language models (LLMs) have expanded the expressive capacity of narrative games by enabling non-player characters (NPCs) to generate context-sensitive dialogue [10]. While such systems can

enhance immersion, they also introduce challenges for narrative coherence, including hallucination and unstable reasoning patterns documented in recent surveys [3, 4].

Investigative narratives place particularly strong demands on coherence. Player reasoning in mystery and investigation genres depends on structured evidence relationships, temporal dependencies, and controlled information disclosure, principles long emphasized in narrative planning and computational storytelling research [11]. Unconstrained generation can violate these requirements by producing statements that contradict established facts or reveal information before they are narratively justified, a failure mode consistent with findings in truthfulness and hallucination detection research [8].

Many existing approaches attempt to address these issues through prompt engineering or reliance on a model’s internal reasoning. However, such methods remain sensitive to prompt formulation and provide limited guarantees of narrative consistency. In addition, standard retrieval-augmented generation (RAG) pipelines integrate external retrieval into language model inference [6], but retrieval is typically driven by semantic similarity rather than explicit narrative constraints. As a result, these approaches provide limited support for enforcing logical dependencies, managing trust relationships, or accounting for spatial conditions within an evolving narrative world. Designers therefore lack mechanisms that make narrative constraints explicit, enforceable, and inspectable, a concern also raised in prior work on authorially controlled interactive systems [2, 7, 9].

This limitation is especially pronounced in investigative game design, where logical fidelity directly shapes player experience. In such settings, narrative truth must be treated as a first-class system component governed by explicit state, rather than as an emergent outcome of language generation.

In response, this paper introduces a constrained generative NPC system grounded in a Dynamic Evidence Graph (DEG). The DEG represents narrative knowledge as interdependent Knowledge Fragments (KFs) whose availability depends on logical dependencies, NPC–player trust levels, spatial context, and the evolving game state. An engine-controlled Eligibility and Gating Funnel determines which fragments may be disclosed at any given moment, producing an Eligible Context Window that constrains generation. The language model operates strictly within this window and serves only as a surface realization layer, while all narrative control remains with the engine.



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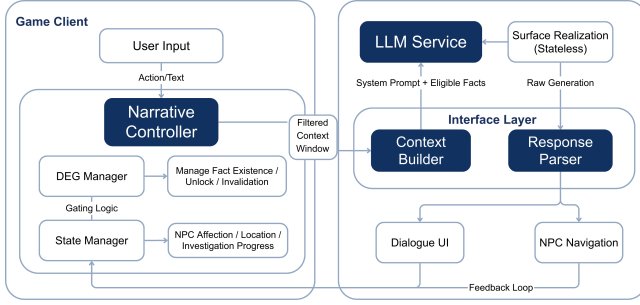


Figure 1: End-to-end architecture of the engine-centric narrative system. Narrative control is handled by a deterministic controller managing ground truth via a Dynamic Evidence Graph, while the language model is used solely for surface realization under constrained context.

We evaluate the system through a 2D investigative game prototype in which players interrogate stationary NPCs, uncover evidence, and submit a final investigative report that determines narrative outcomes. The results indicate that constrained generation, combined with explicit knowledge gating and spatial reasoning, can support coherent investigative gameplay without sacrificing linguistic naturalness.

Beyond the system itself, this work illustrates how explicit evidence structures and engine-controlled knowledge access can support coherent and inspectable generative NPC behavior in investigative games.

2 System Overview

As shown in Figure 1, the system adopts an engine-centric architecture that separates narrative control from language generation. Narrative truth and progression are maintained by the engine through a Dynamic Evidence Graph (DEG), while the language model is used solely for surface realization. Before any response is generated, narrative knowledge is filtered through an Eligibility and Gating mechanism that produces an Eligible Context Window, constraining what NPCs may express.

2.1 Dynamic Evidence Graph

The Dynamic Evidence Graph (DEG) represents narrative knowledge as a set of interdependent Knowledge Fragments (KFs). Each fragment encodes narrative information whose availability depends on prerequisite relations, trust between the NPC and the player, and optional spatial conditions. KFs maintain explicit narrative states that support authoring and runtime verification. As player actions update the DEG, fragments may become available, restricted, or invalidated, allowing the system to revise previously accepted information and prevent contradictions. This structured representation makes narrative truth explicit and inspectable while preserving consistency in LLM-driven dialogue. Although the current prototype does not include a custom visual DEG inspector, narrative state and fragment transitions are observable through Unity’s native Inspector interface for `ScriptableObject` assets and through structured event logs recorded by the runtime system. This provides designers

with access to fragment states, relationship updates, and memory summaries for debugging and authorial oversight.

2.2 Implementation

The prototype was implemented in the Unity game engine using C#. Narrative state is represented through a combination of Unity `ScriptableObject` assets and runtime memory structures. Each NPC is associated with an `NPCData` asset that stores persistent narrative information, including relationship structures (maintained as typed lists), a structured memory summary string used for context construction, and raw conversational memory logs.

The Dynamic Evidence Graph (DEG) is instantiated at runtime as an in-memory structure that maintains fragment dependencies, trust states, and eligibility conditions. Rather than relying on semantic similarity retrieval, fragment eligibility is deterministically recomputed at each dialogue turn.

Large language model responses are generated via the OpenRouter API using an OpenAI-compatible endpoint. After each model response, the system parses structured tags embedded in the output using regular expression matching. These tags trigger real-time updates to internal narrative state, including goal transitions, spatial navigation behavior, and fragment activation. This per-turn gating mechanism ensures that only logically valid Knowledge Fragments are exposed to the model within the current context window.

2.3 Eligibility and Gating

Whether an NPC may disclose a Knowledge Fragment is determined by deterministic eligibility evaluation performed by the engine. Eligibility depends on the fragment’s current state, satisfaction of its dependencies, the NPC’s trust level, and relevant spatial conditions.

At runtime, this evaluation is implemented as an Eligibility and Gating Funnel (Figure 2b), which filters global narrative knowledge into a compact Eligible Context Window. Only fragments satisfying social, spatial, and dependency constraints are exposed for dialogue generation, ensuring that disclosure remains aligned with the current narrative state.

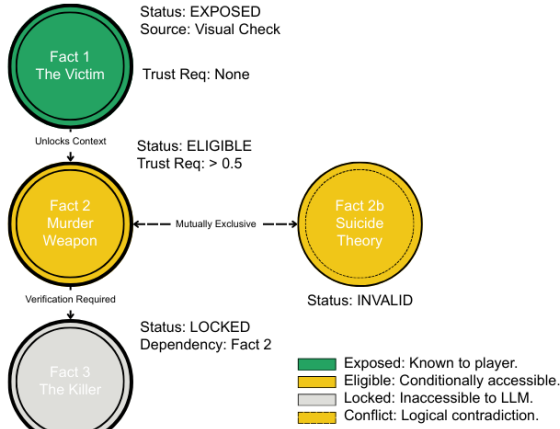
Spatial context is treated as a first-class eligibility parameter. When the player or an NPC enters a new spatial zone, the engine triggers a context update that recomputes fragment eligibility. Spatially bound Knowledge Fragments may therefore become enabled or restricted depending on location, structuring the timing and ordering of evidence disclosure. By integrating spatial constraints directly into eligibility evaluation, the system maintains authorial control while allowing NPC dialogue to adapt to environmental context.

2.4 Constrained Generative Dialogue

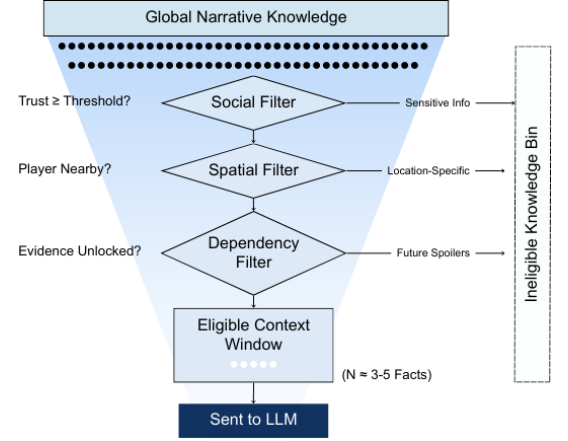
Generative dialogue is produced only after the Eligible Context Window has been constructed. This window contains concise natural language summaries of authorized Knowledge Fragments and serves as the sole narrative input to the language model.

The language model functions purely as a surface realization component and operates strictly within this eligible context, supporting coherent and inspectable NPC behavior without direct access to narrative state or game logic.

Narrative Dependency Scope



(a) Dynamic Evidence Graph structure



(b) Eligibility and gating funnel

Figure 2: Structured truth representation and deterministic eligibility evaluation used to constrain generative dialogue.

2.5 Illustrative Interaction

To illustrate the system in use, we describe a brief interaction from the investigative prototype. The player questions a bartender NPC in a public saloon about whether the murder weapon originated from the bartender’s kitchen.

At this point in the narrative state, the player’s trust level with the bartender is below the required disclosure threshold, and weapon-related Knowledge Fragments remain ineligible due to both social and spatial constraints. The Eligibility and Gating mechanism therefore excludes these fragments from the Eligible Context Window before generation.

The system injects a constraint indicating that the NPC does not yet trust the player sufficiently to reveal sensitive information and limits the response length to maintain conversational pacing. As a result, the language model produces a contextually appropriate deflection, for example:

“You’re asking bold questions for a stranger, partner. I pour whiskey here — not secrets.”

Rather than explicitly refusing to answer, the NPC remains in character while avoiding disclosure. This example demonstrates how deterministic eligibility evaluation constrains generative dialogue and prevents premature revelation of narrative-critical information without sacrificing linguistic naturalness.

3 Formative Expert Feedback

We conducted a formative, perception-based evaluation with four experts in game design and interactive narrative systems. Each session lasted approximately 15 minutes. Participants were asked to interact with the prototype in a predefined investigative scenario in which they attempted to identify the perpetrator of a fictional crime by interrogating NPCs.

Experts were given a short briefing describing the narrative setting but were not constrained to specific dialogue scripts. During

interaction, they were encouraged to explore questioning strategies freely. Following gameplay, we conducted a semi-structured interview focusing on perceived narrative coherence, information control, logical consistency, and the transparency of system behavior.

Feedback indicated that participants perceived the constrained generative approach as improving logical consistency and reducing premature disclosure of critical information. Experts highlighted the clarity of state-dependent responses and noted that eligibility gating contributed to a stronger sense of investigative progression. As this was an early-stage prototype, no comparative baseline condition was included; findings reflect qualitative impressions rather than controlled experimental comparison.

4 Related Work

Interactive narrative research has long examined the tension between player agency and authorial control. Prior work on branching narratives, drama management, and procedural storytelling emphasizes that maintaining coherence in complex story worlds requires explicit structural mechanisms to regulate narrative progression and information disclosure [11]. Investigative game designs impose additional demands, as deductive reasoning depends on ordered evidence, causal dependencies, and controlled information access.

Recent work integrating large language models into games demonstrates their ability to generate fluent, context-sensitive dialogue while also exposing contradictions, hallucinations, and unstable character knowledge [3]. External mechanisms are often required to regulate model outputs. Approaches such as controllable generation [5] and retrieval augmentation [6] can improve reliability, yet they do not enforce explicit narrative state dependencies within interactive systems.

Evidence graphs and dependency-based models have been used to support inference and consistency in investigative storytelling [11, 12], often as authoring or planning tools rather than runtime

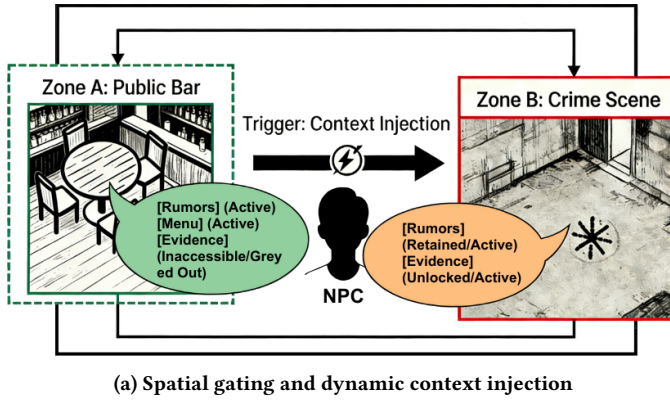


Figure 3: Spatial constraints and their effect on NPC dialogue behavior in the investigative prototype.

controllers. Integrating explicit narrative structures with deterministic gating, as in our system, addresses these limitations by controlling what information is revealed to the model at runtime.

Spatial context has also been shown to influence narrative pacing and access to information. Research on location-based storytelling and embodied agents demonstrates that physical presence and movement shape how players encounter narrative content [1]. However, spatial conditions are rarely treated as explicit constraints in generative dialogue systems. Together, these strands motivate approaches that combine narrative structure, evidence management, and spatial context to regulate NPC dialogue in investigative settings.

5 Conclusion

This work explores a constrained generative NPC system grounded in a Dynamic Evidence Graph (DEG) that separates narrative control from language generation. By representing narrative truth as an explicit and inspectable structure governed by eligibility, trust, and spatial conditions, the system illustrates how generative dialogue can be aligned with the logical requirements of investigative gameplay.

Rather than relying on prompt-level constraints, the approach emphasizes engine-level control over information disclosure, allowing designers to reason about what NPCs may reveal under different narrative conditions. As a poster contribution, this work highlights the potential of structured evidence representations and deterministic gating mechanisms as a foundation for coherent and controllable generative NPC behavior in investigative narratives.

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